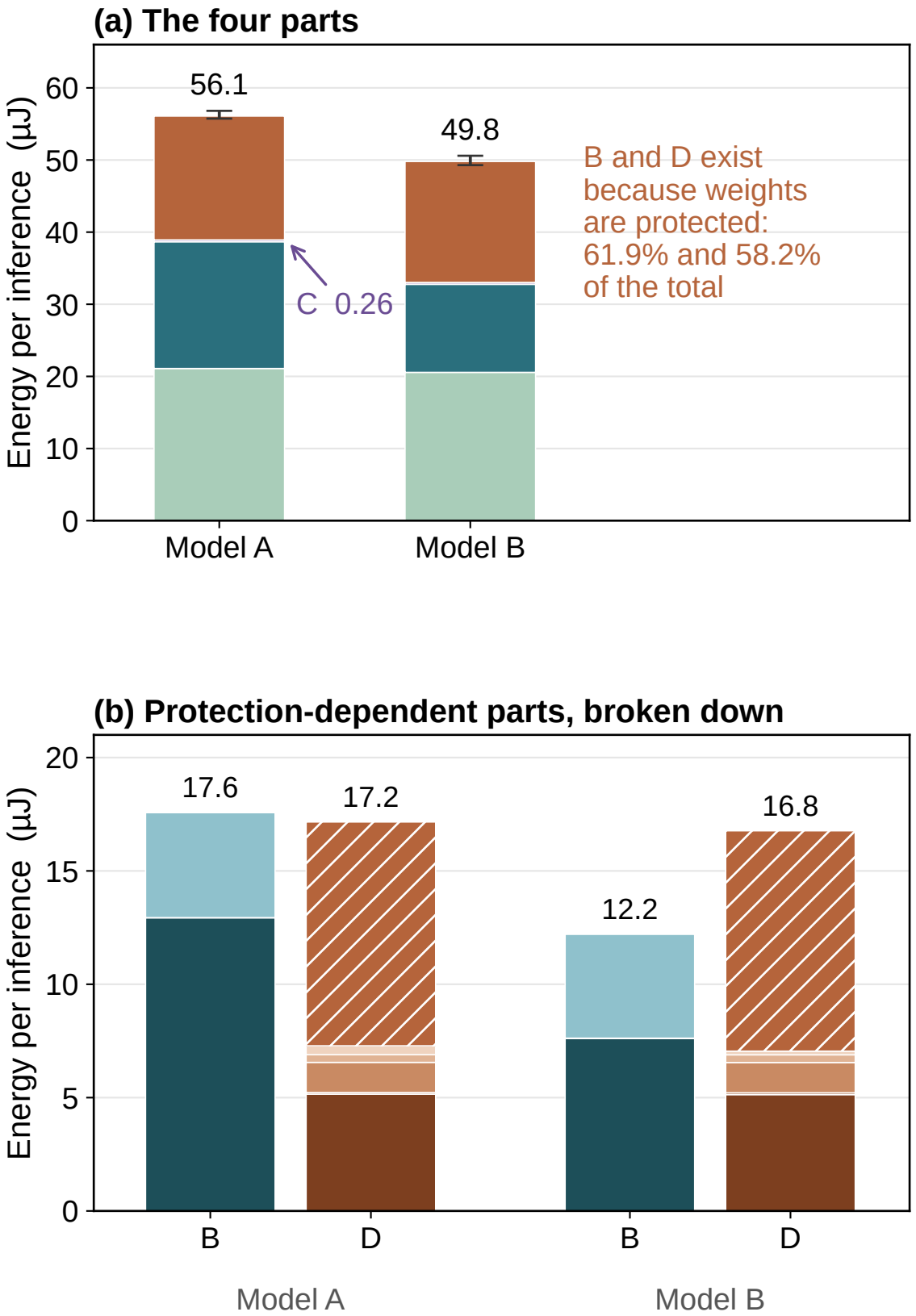
(a) The four parts
Energy per inference (µJ)
0
10
20
30
40
50
60
56.1
49.8
C 0.26
B and D exist because weights are protected: 61.9% and 58.2% of the total
Model A
Model B
(b) Protection-dependent parts, broken down
Energy per inference (µJ)
0
5
10
15
20
17.6
17.2
12.2
16.8
B
D
B
D
Model A
Model B
A digital periphery
B protected-weight storage
C PCM-CIM array
D address map
B: SRAM reads
B: conv1 cache
D: ROM lookup reads
D: ROM parking reads
D: static after ROM lookup reads
D: ROM at rest, all-ones word
D: ROM parked on zero bits
D: AMU logic